

Vocabulary The speed of change; describing cause and effect

1 Work in pairs and discuss these questions.

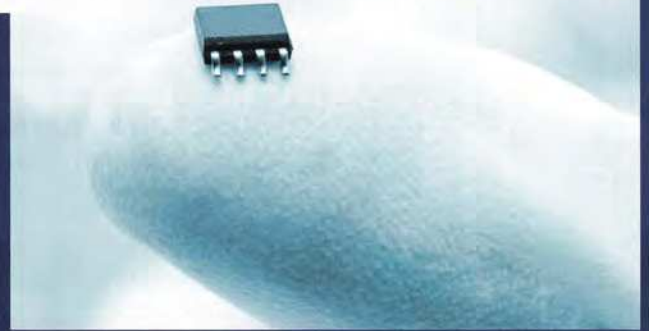
- 1 In what ways has computer technology changed our lives? Make a list.
- 2 Do you think the world is becoming more competitive? Why / Why not?

2 Read the text and answer the questions below.

The power of numbers

There is an Indian legend that goes like this: once upon a time, a king challenged a pilgrim to a game of chess. The pilgrim agreed, but on one condition: if the king lost, he would put a single grain of rice on the first chess square and then double the amount on every square after that. The king lost and then discovered the price he had to pay: 1 grain on the first square, 2 on the second, 4 on the third, 8 on the fourth, and so on. By the 64th square, the amount came to more than 18,000,000,000,000,000,000 grains of rice!

A modern example of this type of growth is Moore's Law. This law says that the power of computer microchips doubles every 18 months. We can see the effects of this if we compare the space shuttle with a smartphone. The computer on the first space shuttle could process 400,000 instructions per second and had a memory of 424 kilobytes. A modern smartphone is over 35,000 times quicker and has over 33,000,000 times more memory! These increases in computing power have caused scientific discovery to speed up. It took the Human Genome Project 13 years, from 1990 to 2003, to analyse the entire human genome*. Due to faster computers, scientists can now do this in a week. In the next decades, computers will equal the power of the human brain. This may result in the development of computers that are more intelligent than humans. The possible consequences of this are profound.



The increasing speed of scientific discovery is also partly the result of larger social trends. In the 1960s in the UK, about 5% of young people went to university. Nowadays, 40% of people in the UK do a degree, whilst in Finland the number has reached 80%. Between 2000 and 2010, the number of students studying in universities abroad rose by 75%. Because of the increase in university students, more people now do research at a postgraduate level. This has created greater competition between research departments for funding. As a result, there is more pressure on scientists to publish their findings quickly. This new research leads to other new discoveries and, as a consequence, new technologies are developed more rapidly.

The speed of change is hard to cope with because it means we need to keep learning and adapting. But we'd better get used to it: it's not going to slow down!



*genome = the complete set of genes in a cell or living thing

- 1 Why did the king have to give the pilgrim so much rice?
- 2 What is Moore's Law?
- 3 Which has more memory: the computer on the first space shuttle or a smartphone?
- 4 Why is it now quicker to analyse human genes?
- 5 How many people now go to university in the UK and in Finland?
- 6 Why is research now published more quickly?